

House Price Prediction – Summary

This project focused on predicting house prices using machine learning techniques and analyzing the factors that have the greatest impact on property values. The housing dataset was first explored and cleaned by checking for missing values, removing duplicate records, and converting categorical features into numerical format. Exploratory data analysis was then performed to better understand the distribution of house prices and the relationships between different features.

Two regression models were developed and evaluated: Linear Regression and Random Forest Regressor. The dataset was split into training and testing sets using an 80:20 ratio. Model performance was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. The Linear Regression model achieved an MAE of 970,043, an RMSE of 1,324,507, and an R^2 score of 0.653. The Random Forest model achieved an MAE of 1,021,546, an RMSE of 1,400,566, and an R^2 score of 0.612. Based on these results, Linear Regression performed slightly better and was selected as the more effective model for this dataset.

Feature importance analysis revealed that the area of a house was the most significant factor influencing its price, contributing nearly half of the model's predictive power. Other important features included the number of bathrooms, availability of air conditioning, parking spaces, number of stories, and bedrooms. The correlation analysis also showed positive relationships between these features and house prices, indicating that larger homes with more amenities generally command higher market values.

One interesting observation was that the simpler Linear Regression model outperformed the more complex Random Forest model. This suggests that the relationships within the dataset are relatively linear and can be captured effectively without requiring a highly complex model. Overall, the project demonstrated how machine learning can be used to estimate house prices and identify the key factors that drive property values. These findings can help real estate businesses make more informed pricing decisions and better understand the characteristics that contribute most to a property's market value.